Sardar Patel Institute of Technology
Major Project Phase-1

TerraFlow BioAI

Closed-Loop Embedded Precision Irrigation & Environmental Intelligence

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The Problem

What smallholder farmers face every day



Schedule-Based Irrigation

Farmers water crops on fixed schedules, not on actual soil demand — leading to overwatering or drought stress.



Reactive Disease Detection

Disease is detected only after visible damage appears. There is no early warning from environmental conditions.



No Real-Time Soil Data

Zero visibility into soil moisture, temperature, or salinity. Decisions rely purely on guesswork and experience.



No Trend Monitoring

No historical data means farmers cannot identify patterns or optimize irrigation over time.

System Architecture

End-to-end data flow from sensors to dashboard



1	**Sensors → STM32**	5 sensors continuously acquired via ADC, I2C, OneWire. STM32 filters data and runs local decision logic.
2	**Local Control Loop**	STM32 evaluates moisture + humidity + temperature thresholds. Activates MOSFET → pump if stress detected. Validates result.
3	**Wireless Telemetry**	STM32 serializes JSON over UART to ESP-01S. ESP sends HTTP POST to Node.js backend over Wi-Fi.
4	**Backend Processing**	Node.js computes WSI, DRI indices. Calls Python ML microservice. Stores everything in Supabase PostgreSQL.
5	**Dashboard + AI**	React dashboard polls /latest. Displays live sensor cards, charts, agro-indices, pump state, and ML irrigation prediction.

5-Sensor Stack

Soil + Environmental multi-modal sensing architecture



Capacitive Moisture

PA0 ADC • Soil moisture %

Primary irrigation trigger. Capacitive — no electrolysis, no corrosion.



DS18B20 Waterproof

PA4 OneWire • Soil temperature °C

TDS compensation + root-zone thermal context for ML model.



Analog TDS Sensor

PA1 ADC • Dissolved solids ppm

Salinity/fertilizer proxy. Detects osmotic stress at >1200 ppm.



BME280 (HW-611)

PB6/7 I2C • Temp + Humidity + Pressure

VPD & DRI computation. Three parameters from one I2C chip.



BH1750 Light Sensor

PB6/7 I2C • Illuminance (lux)

PAR proxy. Radiation load feeds WSI. Shared I2C bus with BME280.

Hardware Core

STM32F103C8T6 — Controller, power architecture, and actuation circuit

STM32F103C8T6 Blue Pill

- ARM Cortex-M3 @ 72 MHz
- 12-bit ADC — 0 to 4095 range
- Hardware USART1 + USART2
- I2C on PB6/PB7 — shared bus
- 3 GPIO PWM for pump control
- Local decision engine — no cloud dependency

MOSFET Actuation Circuit

- IRLZ44N — logic-level MOSFET
- 220Ω gate resistor from PB0
- 10kΩ pull-down — safe default OFF
- 1N4007 flyback diode across pump
- Silent, fast, no contact wear
- Water tank → pump → tube → soil

Power Architecture

5V 2A Adapter → **LM2596 Buck** → **3.3V Rail** → **STM32 + BME280 + BH1750 + TDS + Moisture Sensors**

Agro-Intelligence Indices

WSI — Water Stress Index • DRI — Disease Risk Index

Water Stress Index (WSI)

0 – 100 score. Combines 3 stress signals into one irrigable metric.

$$WSI = (0.5 \times moisture_stress) + (0.3 \times VPD_norm) + (0.2 \times lux_norm) \times 100$$

Moisture Stress	*1 − (moisture / 100)*	**50%**
VPD Norm	*min(VPD / 3.0, 1.0)*	**30%**
Lux Norm	*min(lux / 80000, 1.0)*	**20%**

0–30	30–60	60–80	80–100
No Stress	Mild Stress	High Stress	Critical

Sensors: Moisture + BME280 (temp/hum) + BH1750 (lux)

Disease Risk Index (DRI)

0 – 100 score. Tracks fungal/bacterial outbreak risk from humidity and temperature.

$$DRI = humidity_factor \times temp_suitability \times 100$$

Temp Suitability	1.0 if 15–30°C, 0.5 if 10–35°C, 0.1 outside
Humidity Factor	0 below 60% RH → linear ramp → 1.0 above 80%
Pathogen Basis	Models powdery mildew, botrytis, bacterial leaf spot
No leaf wetness sensor needed	RH >80% is accurate proxy for leaf surface moisture

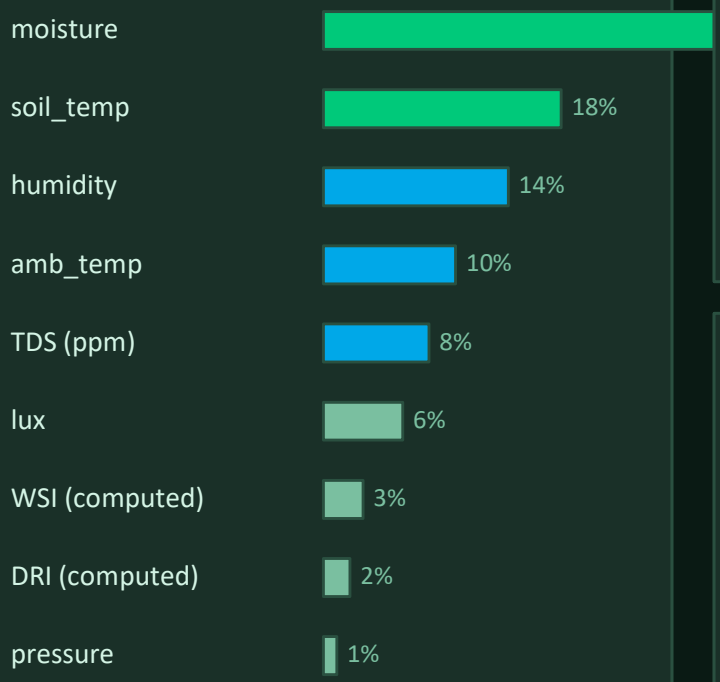
0–30 Low Risk	30–60 Moderate	60–100 High Risk

Sensors: BME280 only (ambient temp + relative humidity)

ML-Assisted Irrigation Prediction

Random Forest classifier — decision support layer, not primary controller

9 Input Features



Model Details

Type:	Random Forest
Output:	IRRIGATE / NO
Also:	Confidence %
Service:	Python Flask
Library:	scikit-learn

Positioning

- ✓ STM32 = primary controller
- ✓ ML = decision support only
- ✓ Safer & more realistic
- ✓ AI assists — doesn't command

Prediction Flow

STM32 — Sensor readings acquired locally

↓

ESP-01S — JSON over UART → Wi-Fi HTTP

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Node.js — Computes WSI, DRI Calls ML service

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Python RF — Classifies: IRRIGATE or NO IRRIGATE

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Dashboard — Shows prediction + confidence %

Crop Recommendation System

✓ MODEL DEPLOYED

ML-powered crop suitability scoring — model deployed in production

How It Works

1 Live Sensor Inputs

Moisture, soil temp, TDS, humidity, ambient temp, pressure, and lux are read continuously from the 5-sensor stack.

2 WSI & DRI Computed

Water Stress Index and Disease Risk Index are calculated from sensor data on the Node.js backend.

3 9 Features Assembled

All sensor values + computed indices are packaged as a feature vector — same pipeline as the irrigation ML model.

4 Crop Scorer Runs

Each candidate crop is scored against its ideal parameter ranges. Multi-condition weighted scoring function.

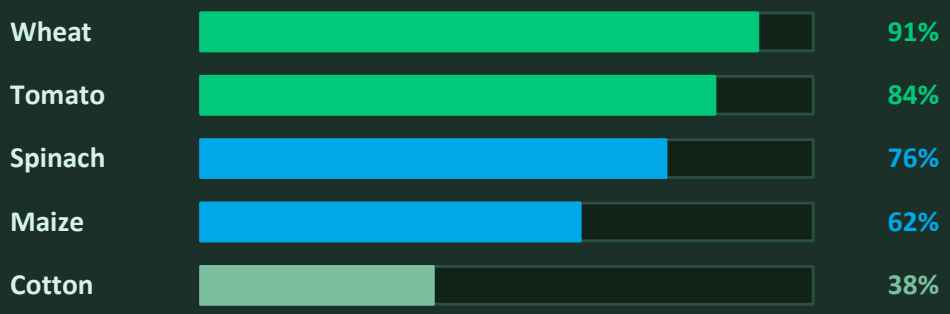
5 Top Crops Returned

Top 3–5 crops ranked by suitability score. Stored as JSONB in Supabase, displayed live on dashboard.

Input Features

▸ **Moisture %**	Soil moisture level	▸ **Soil Temp °C**	Root zone temperature
▸ **TDS ppm**	Nutrient / salinity proxy	▸ **Humidity %**	Ambient relative humidity
▸ **Lux**	Light availability (PAR proxy)	▸ **WSI**	Computed water stress index
▸ **DRI**	Computed disease risk index	▸ **Pressure hPa**	Barometric / altitude

Sample Recommendations (Live Output)



Stored as JSONB in Supabase • Live on React dashboard

Backend & Cloud Storage

Node.js Express • Supabase PostgreSQL • React Dashboard

Node.js API Routes

POST	/sensor-data	Receive JSON from ESP-01S. Compute WSI, DRI. Store.
GET	/latest	Return most recent reading for dashboard poll.
GET	/history	Return time-series for trend charts.
POST	/pump	Dashboard manual pump override command.
GET	/predict	Proxy call to Python ML Flask service.

Supabase — TerraFlowBioAI Table

moisture, soil_temp, tds	Core sensor readings
amb_temp, humidity, pressure	BME280 environmental
lux	BH1750 light
vpd, wsi, dri	Computed agro-indices
shi, mai	Future ESE-2 indices
prediction, confidence	ML output
crops (JSONB)	Crop recommendation
pump_state, irrigation_effective	System state

React Dashboard Components

Moisture Trend Chart | WSI Gauge | DRI Gauge | 5 Sensor Cards | ML Prediction Panel | Pump Control Buttons | Alert Feed

Closed-Loop Demo Sequence

Live demonstration of the complete autonomous irrigation loop

1 Pull Moisture Probe from Soil

Moisture ADC value rises toward dry reference. Moisture % drops on dashboard.

2 WSI Climbs in Real Time

Moisture stress component increases. VPD and lux contribute. WSI crosses 60 → High Stress (orange).

3 Local Logic Fires

STM32 detects moisture below 35% threshold. Local decision engine triggers independently of cloud.

4 ML Prediction Flips

Python Random Forest returns IRRIGATE with confidence value. Dashboard AI panel updates.

5 MOSFET Activates — Pump Runs

PB0 goes HIGH. IRLZ44N switches. Water flows from tank through tube to soil. pump_state = TRUE.

6 Moisture Recovers — Pump Stops

STM32 re-reads moisture. Threshold crossed. PB0 LOW. irrigation_effective = TRUE stored in DB.

ESE-2 Roadmap

Planned upgrades — hardware, models, and indices

Hardware

- → Rain sensor FC-37 — suppress pump during rainfall (₹80)
- → 2nd capacitive moisture — spatial variability proof
- → MLX90614 — canopy IR temperature
- → MH-Z19B CO_2 — soil biological activity proxy

ML Upgrade

- → LSTM replacing Random Forest
- → Trained on real collected time-series data
- → Sequence-aware irrigation scheduling

New Indices

- → SHI — Soil Health Index (TDS + DS18B20 + CO_2)
- → MAI — Microbial Activity Index
- → ET_o estimation with barometric pressure
- → VPD-based crop-specific thresholds

Infrastructure

- → WebSocket real-time instead of polling
- → Supabase row-level security
- → Alert email/SMS notifications
- → Offline STM32 SD-card data buffering

System Summary

5 Sensors

Moisture • Temp • TDS • BME280 • BH1750

STM32 Core

72MHz ARM • ADC • I2C • OneWire • UART

Autonomous Loop

Local threshold + feedback validation

WSI + DRI

Water Stress & Disease Risk Indices

Random Forest

9-feature ML irrigation classifier

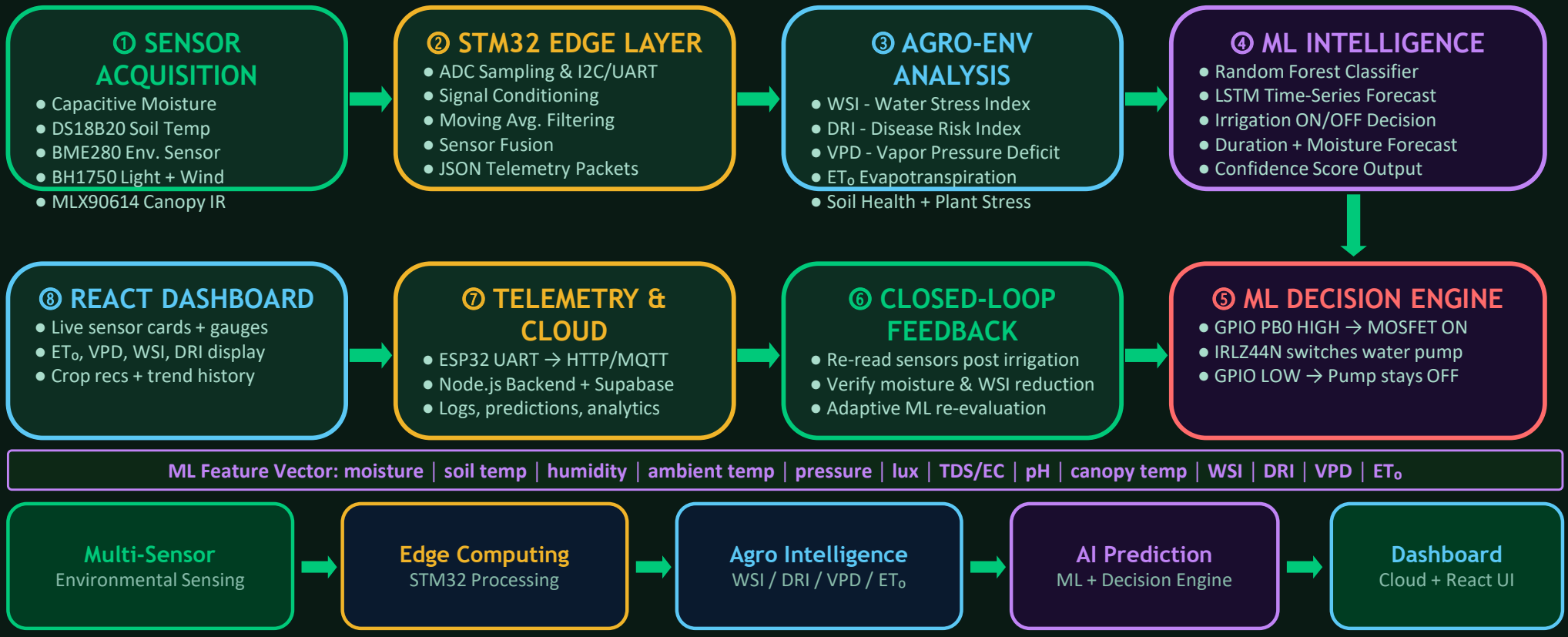
Cloud + Dashboard

Supabase • Node.js • React live UI

"A real embedded system that senses soil and environment continuously, irrigates automatically when stress is detected, and provides live intelligent dashboard with ML-assisted prediction and agro-index monitoring."

System Workflow

AI-driven closed-loop precision irrigation: from sensor acquisition to intelligent actuation



ML Feature Vector: moisture | soil temp | humidity | ambient temp | pressure | lux | TDS/EC | pH | canopy temp | WSI | DRI | VPD | ET_0